Siddhant Ray

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Education

The University of Chicago 2023 – Present
PhD in Computer Science
Advisor - Junchen Jiang and Nick Feamster

ETH Zürich 2020 – 2022
MSc in Electrical Engineering and Information Technology
Advisor - Laurent Vanbever

VIT Vellore 2016 – 2020
B.Tech in Electronics and Communication Engineering

Experience

Computer Science Department, The University of Chicago Sep 2023 – Present
Graduate Research Assistant

- Joint quality-cost optimization for LLM Agentic Systems with unified KV Cache management and application feedback.
- Joint quality-latency optimization for Retrieval Augmented Generation (RAG) systems in LLM inference.
- Transformer models for packet-level queue management and tail-latency reduction in latency-sensitive L4S applications.

Microsoft Research Jun 2026 – Sep 2026
Research Intern

- Developed a Mixture-of-ActionHeads (MoA) VLA model for efficient, dynamic inference on long-horizon physical AI tasks.
- Built co-affinity based GPU-CPU offloading across action heads, increasing task throughput by 6X on the LIBERO benchmark.
- Jointly optimized task-aware scheduling and head management for further improving task throughput by 2X.

Microsoft Research Jun 2025 – Sep 2025
Research Intern

- Joint internship with Outlook for scalable and cost-efficient LM architectures for email importance labeling.
- Built a hybrid architecture with SLM cascades and classifiers to reduce labeling COGS by 150X with negligible quality loss.
- Email labeling system has been deployed in Outlook's production email client.

Advanced Network Architectures Lab, UPC Barcelona Sep 2022 – Mar 2023
Cloud Networks Researcher

- Developed an approximation for Mixed-Integer Optimal Matching to reduce execution time by 2.5-3X on edge VM servers.

Law, Economics, and Data Science Group, ETH Zurich Oct 2021 – Sep 2022
Graduate Research Assistant

- Improved semantic labeling pipelines for sentence simplification, paraphrase mining, and topic clustering.
- Developed NLP models to automatically analyse discourse in policy documents for actor-narrative clustering.

Capgemini Engineering May 2019 – July 2019
Software Development Intern

- Developed a K-Shortest Path Searching algorithm in an ONOS based Software Defined Layer 2 VPN.
- Algorithm applied dynamic constraints of network resources (e.g., mandatory edges) for path calculation.

BlueStacks May 2018 – July 2018
Software Development Intern

- Developed an ML algorithm to customize the Bluestack's display screen using past users' experiences.
- Built an automation tool to create SVG cards and an address verification tool for Bluestack's game engine.

Publications

Argo: Efficient Importance Labeling for Enterprise Email Systems. 2026
Siddhant Ray, Ganesh Ananthanarayanan, Kevin Chian, Yan Guo, Cristina St Hill, Jack W. Stokes, Victor Wang, and Junchen Jiang.
5th Workshop on Practical Adoption Challenges of ML for Systems (PACMI @ SOSP'26).

SwiftQueue: Optimizing Low-Latency Applications with Swift Packet Queuing. 2026
Siddhant Ray, Xi Jiang, Jack Luo, Nick Feamster, and Junchen Jiang.
New Ideas in Networked Systems (NINeS'26).

METIS: Fast Quality-Aware RAG Systems with Configuration Adaptation. 2025
Siddhant Ray, Rui Pan, Zhuohan Gu, Kuntai Du, Shaoting Feng, Ganesh Ananthanarayanan, Ravi Netravali, and Junchen Jiang.
ACM SIGOPS 31st Symposium on Operating Systems Principles (SOSP'25).

CacheBlend: Fast Large Language Model Serving for RAG with Cached Knowledge Fusion. 2025
Jiayi Yao, Hanchen Li, Yuhan Liu, Siddhant Ray, Yihua Cheng, Qizheng Zhang, Kuntai Du, Shan Lu, and Junchen Jiang.
ACM European Conference on Computer Systems (EuroSys'25).

CacheGen: KV Cache Compression and Streaming for Fast Large Language Model Serving. 2024
Yuhan Liu, Hanchen Li, Yihua Cheng, Siddhant Ray, Yuyang Huang, Qizheng Zhang, Kuntai Du, Jiayi Yao, Shan Lu, Ganesh Ananthanarayanan, Michael Maire, Henry Hoffmann, Ari Holtzman, and Junchen Jiang.
ACM Special Interest Group on Data Communication (SIGCOMM'2024).

Eloquent: A More Robust Transmission Scheme for LLM Token Streaming. 2024
Hanchen Li, Yuhan Liu, Yihua Cheng, Siddhant Ray, Kuntai Du, and Junchen Jiang.
SIGCOMM Workshop on Networks for AI Computing (NAIC @ SIGCOMM'24).

A New Hope for Network Model Generalization. 2022
Alexander Dietmüller, Siddhant Ray, Romain Jacob, and Laurent Vanbever.
21st ACM Workshop on Hot Topics in Networks (Hotnets'22).

Machine Learning based Cell Association for mMTC 5G Communication Networks. 2020
Siddhant Ray and Budhaditya Bhattacharyya.
International Journal of Mobile Network Design and Innovation.

Honors and Awards

I-Corps Cohort, *Polsky Center and Booth School*, University of Chicago 2026
Travel Grant, *LDOS PhD Research School*, UT Austin 2025
Best Paper Award, *EuroSys 2025*, Rotterdam 2025
Liew Family Graduate Fellowship, University of Chicago 2023 – 2028
Winner at Datathon, *Microsoft Challenge*, ETH Zurich 2022
Best Outgoing Student, *SENSE department*, VIT Vellore 2020
Merit Scholarship for Academic Excellence, VIT Vellore 2016 – 2019

Patents

Document Labeling With Language Model Cascade, *Microsoft*, Pending Mar 2026

Talks

Seminar Talk, *NAISS Lab Seminar*, HKU April 2026
Seminar Talk, *Systems Research Seminar*, UIUC Nov 2025
Guest Lecture, *Systems for LLMs and AI Agents Class*, UCSD Nov 2025
PhD Research Talk, *Networking Research Group*, Microsoft Research Aug 2025
Lightning Talk, *Research Social*, Conviva April 2024

Service

Reviewer, *AAAI, ICLR* 2027
External Review Committee, *MLSys* 2026
Reviewer, *AAAI, ICLR, NeurIPS* 2026
Reviewer, *ICML* 2025
Artifact Evaluation Committee, *ATC, OSDI, CoNEXT* 2025
Artifact Evaluation Committee, *CoNEXT* 2024

Teaching

Teaching Assistant, *Systems Programming II*, UChicago 2026
Teaching Assistant, *Systems Programming II*, UChicago 2025

Other Research Projects

Advancing Packet-Level Traffic Predictions with Transformers (Master Thesis) - [code, thesis] 2022
Towards a New Framework for Integration of Network Planes (Research Project) - [code] 2021

Technical Skills

Programming Python, C++, C, Bash, Rust, SQL, Java, TEX
Software Linux, Git, Docker, P4, Azure, Google Cloud, AWS, Maven, K8s, Helm
Frameworks vLLM, Langchain, Llama-Index, PyTorch, Sklearn, NLTK, Flask, Asyncio, NS3, Mininet, FRRouting